APM1220 FA1

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Problem 2.2 **(5 points)**

Given the matrices

$$\mathbf{A} = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{C} = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}$$

perform the indicated multiplications:

(a) $5\mathbf{A}$

$$5A = 5\begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} -1 \cdot 5 & 3 \cdot 5 \\ 4 \cdot 5 & 2 \cdot 5 \end{bmatrix}$$

$$\boxed{5A = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}}$$

(b) $\mathbf{BA}$

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4(-1) + (-3)(4) & 4(3) + (-3)(2) \\ 1(-1) + (-2)(4) & 1(3) + (-2)(2) \\ (-2)(-1) + (0)(4) & (-2)(3) + (0)(2) \end{bmatrix}$$

$$= \begin{bmatrix} -4 - 12 & 12 - 6 \\ -1 - 8 & 3 - 4 \\ 2 + 0 & -6 + 0 \end{bmatrix}$$

$$\boxed{BA = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}}$$

(c) $\mathbf{A'B'}$

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \qquad B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (-1)(4) + (4)(-3) & (-1)(1) + (4)(-2) & (-1)(-2) + (4)(0) \\ (3)(4) + (2)(-3) & (3)(1) + (2)(-2) & (3)(-2) + (2)(0) \end{bmatrix}$$

$$= \begin{bmatrix} -4 - 12 & -1 - 8 & 2 + 0 \\ 12 - 6 & 3 - 4 & -6 + 0 \end{bmatrix}$$

$$\boxed{A'B' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}}$$

(d) $\mathbf{C'B}$

$$C' = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix}$$

$$C'B = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (5)(4) + (-4)(1) + (2)(-2) & (5)(-3) + (-4)(-2) + (2)(0) \end{bmatrix}$$

$$= \begin{bmatrix} 20 - 4 - 4 & -15 + 8 + 0 \end{bmatrix}$$

$$\boxed{C'B = \begin{bmatrix} 12 & -7 \end{bmatrix}}$$

(e) Is **AB** defined?

For a matrix multiplication to be defined, the number of columns of the 1st matrix must equal the number of rows of the 2nd one.

The matrix **A** has dimensions 2×2, while the matrix **B** has dimensions 3×2. **BA** is defined because the number of columns in **B** and the number of rows in **A** are both 2.

Similarly, $\mathbf{A}'\mathbf{B}'$ is defined because the number of columns in $\mathbf{A}'$ is 2 and the number of rows in $\mathbf{B}'$ is also 2.

For the multiplication $\mathbf{AB}$ to be defined, the number of columns of $\mathbf{A}$ must equal the number of rows of $\mathbf{B}$. The number of columns of $\mathbf{A}$ is 2, while the number of rows of $\mathbf{B}$ is 3. Since

$$2 \neq 3,$$

the multiplication is not defined.

$$\boxed{\mathbf{AB} \text{ is not defined.}}$$

Problem 2.4 **(6 points)**

When $\mathbf{A}^{-1}$ and $\mathbf{B}^{-1}$ exist, prove each of the following:

(a) $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$

Since $\mathbf{A}^{-1}$ exists, by definition of an inverse we know that

$$\mathbf{AA}^{-1} = \mathbf{I}.$$

Taking the transpose of both sides gives

$$(\mathbf{AA}^{-1})' = \mathbf{I}'.$$

Since the transpose of an identity matrix is itself, $\mathbf{I}' = \mathbf{I}$. Also, the transpose of a product reverses the order of the matrices, so

$$\underline{(\mathbf{A}^{-1})'\mathbf{A}' = \mathbf{I}}$$

Similarly, using the other side of the inverse definition,

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I},$$

taking the transpose of both sides will result in

$$(\mathbf{A}^{-1}\mathbf{A})' = \mathbf{I}',$$

and applying the same properties as before will give us

$$\underline{\mathbf{A}'(\mathbf{A}^{-1})' = \mathbf{I}}$$

We showed that $(\mathbf{A}^{-1})'$ produces the identity matrix $\mathbf{I}$ when multiplied by $\mathbf{A}'$ on both the left and the right. By definition, it is the inverse of $\mathbf{A}'$. Therefore,

$$\boxed{(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'}$$

(b) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

Consider the product

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}).$$

Using associativity property of matrix multiplication, we get

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B}.$$

Since $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$,

$$\mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{IB}.$$

Since multiplying by the identity matrix simply results in the original matrix, we have

$$\mathbf{B}^{-1}\mathbf{IB} = \mathbf{B}^{-1}\mathbf{B}$$

By definition of an inverse we know that $\mathbf{B}^{-1}\mathbf{B} = \mathbf{I}$

Therefore,

$$\underline{(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{I}}$$

Similarly,

$$(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1}$$

Since $\mathbf{B}^{-1}\mathbf{B} = \mathbf{I}$,

$$(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{AIA}^{-1}$$

Since multiplying by the identity matrix simply results in the original matrix, we have

$$\mathbf{A}^{-1}\mathbf{IA} = \mathbf{A}^{-1}\mathbf{A}$$

By definition of an inverse we know that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$

Therefore,

$$\underline{(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}}$$

Since $\mathbf{B}^{-1}\mathbf{A}^{-1}$ produces the identity matrix $\mathbf{I}$ when multiplied by $\mathbf{AB}$ on both the left and the right, it is, by definition, the inverse of $\mathbf{AB}$. Therefore,

$$\boxed{(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}}$$

Problem 2.8 **(6 points)**

Given the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

find the eigenvalues λ_1 and λ_2 and the associated normalized eigenvectors $\mathbf{e}_1$ and $\mathbf{e}_2$. Determine the spectral decomposition of $\mathbf{A}$.

Finding the eigenvalues
The eigenvalues of matrix $\mathbf{A}$ can be found by solving the equation: $det(\lambda\mathbf{I} - \mathbf{A})$

$$det\left(\lambda\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}\right) = 0$$

$$det\left(\begin{bmatrix} \lambda - 1 & -2 \\ -2 & \lambda + 2 \end{bmatrix}\right) = 0$$

Computing the determinant,

$$(\lambda - 1)(\lambda + 2) - (-2)(-2) = 0$$

$$\lambda^2 + \lambda - 2 - 4 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

Factoring the quadratic gives

$$(\lambda - 2)(\lambda + 3) = 0$$

Therefore, the eigenvalues are

$$\boxed{\lambda_1 = 2, \qquad \lambda_2 = -3}$$

To solve for the eigenvectors, the equation $(\mathbf{A} - \lambda \mathbf{I})e = 0$ will be used.

Finding the eigenvector for $\lambda_1 = 2$:
We solve

$$(\mathbf{A} - 2\mathbf{I})\mathbf{e}_1 = \mathbf{0}.$$

$$\begin{bmatrix} 1-2 & 2 \\ 2 & -2-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row,

$$-x + 2y = 0$$

$$x = 2y.$$

Let $y = 1$. Then $x = 2$, so an eigenvector is

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Its length is

$$\sqrt{2^2 + 1^2} = \sqrt{5}.$$

Therefore, the normalized eigenvector is

$$\boxed{\mathbf{e}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}}$$

Finding the eigenvector for $\lambda_2 = -3$:
We solve

$$(\mathbf{A} + 3\mathbf{I})\mathbf{e}_2 = \mathbf{0}.$$

$$\begin{bmatrix} 1+3 & 2 \\ 2 & -2+3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row,

$$4x + 2y = 0$$

$$y = -2x.$$

Let $x = 1$. Then $y = -2$, so an eigenvector is

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Its length is

$$\sqrt{1^2 + (-2)^2} = \sqrt{5}.$$

Therefore, the normalized eigenvector is

$$\boxed{\mathbf{e}_2 = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}}$$

Spectral decomposition
The spectral decomposition (2-16) is

$$\mathbf{A} = \lambda_1 \mathbf{e}_1 \mathbf{e}_1' + \lambda_2 \mathbf{e}_2 \mathbf{e}_2'.$$

$$\mathbf{e}_1 \mathbf{e}_1' = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\mathbf{e}_2 \mathbf{e}_2' = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$$

$$\mathbf{A} = 2 \cdot \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} + (-3) \cdot \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 8/5 & 4/5 \\ 4/5 & 2/5 \end{bmatrix} + \begin{bmatrix} -3/5 & 6/5 \\ 6/5 & -12/5 \end{bmatrix}$$

$$\boxed{\mathbf{A} = 2 \left(\frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \right) + (-3) \left(\frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \right) = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}}$$

which recovers $\mathbf{A}$, confirming the decomposition is correct.

Problem 2.11 **(5 points)**

Show that the determinant of the $p \times p$ diagonal matrix $\mathbf{A} = \{a_{ij}\}$ with $a_{ij} = 0, i \neq j$, is given by the product of the diagonal elements; thus, $|\mathbf{A}| = a_{11} a_{22} \cdots a_{pp}$.

Proof
Starting with cofactor expansion along the first row, using Definition 2A.24:

$$|\mathbf{A}| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1p}A_{1p}$$

where A_{1j} is the cofactor of a_{1j}. Since $\mathbf{A}$ is diagonal, every term in the first row is zero except the first term. Thus

$$|\mathbf{A}| = a_{11}A_{11} + 0 + \cdots + 0$$

$$|\mathbf{A}| = a_{11}A_{11}$$

$\mathbf{A}_{11}$, is also the submatrix obtained by deleting the first row and first column of $\mathbf{A}$. Because $\mathbf{A}$ is diagonal, $\mathbf{A}_{11}$ is itself a diagonal matrix, with diagonal entries $a_{22}, a_{33}, \ldots, a_{pp}$:

$$\mathbf{A}_{11} = \begin{bmatrix} a_{22} & 0 & \cdots & 0 \\ 0 & a_{33} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & a_{pp} \end{bmatrix}$$

Again if we expand $|\mathbf{A}_{11}|$ along its first row, which again has only one nonzero entry, a_{22}, this gives us

$$\mathbf{A}_{11} = a_{22}\mathbf{A}_{22}$$

Thus,

$$|\mathbf{A}| = a_{11}a_{22}A_{22}$$

Continuing this process, we will obtain, after $p-1$ reductions,

$$|\mathbf{A}| = a_{11} \cdot a_{22} \cdot \mathbf{A}_{22} = a_{11}a_{22}a_{33} \cdots a_{p-1,p-1} \cdot a_{pp}.$$

Therefore,

$$\boxed{|\mathbf{A}| = a_{11}a_{22} \cdots a_{pp}}$$

Problem 2.24 **(5 points)**

Let $\mathbf{X}$ have covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Find:

(a) $\mathbf{\Sigma}^{-1}$

Since $\mathbf{\Sigma}$ is a diagonal matrix, its inverse is also a diagonal matrix. The diagonal entries of the inverse can simply be obtained by taking the reciprocals of the entries of the original matrix, thus,

$$\boxed{\mathbf{\Sigma}^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

(b) The eigenvalues and eigenvectors of $\mathbf{\Sigma}$

For a diagonal matrix such as $\mathbf{\Sigma}$, the eigenvalues λ_i are simply the diagonal entries d_{ii}, and the eigenvectors e_i are simply the corresponding basis vectors where the i-th component is 1 and the rest is zero. Thus,

$$\boxed{\lambda_1 = 4,\ \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \qquad \lambda_2 = 9,\ \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \qquad \lambda_3 = 1,\ \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}$$

(c) The eigenvalues and eigenvectors of $\mathbf{\Sigma}^{-1}$

Again since $\mathbf{\Sigma}^{-1}$ is also a diagonal matrix, its the eigenvalues λ_i are simply the diagonal entries d_{ii}, and the eigenvectors e_i are simply the corresponding basis vectors where the i-th component is 1 and the rest is zero. Thus,

$$\boxed{\lambda_1 = \frac{1}{4},\ \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \qquad \lambda_2 = \frac{1}{9},\ \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \qquad \lambda_3 = 1,\ \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}$$

Problem 2.30 **(8 points)**

You are given the random vector $\mathbf{X}' = [X_1, X_2, X_3, X_4]$ with mean vector $\boldsymbol{\mu}'_X = [4, 3, 2, 1]$ and variance-covariance matrix

$$\mathbf{\Sigma}_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}$$

Partition $\mathbf{X}$ as $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{bmatrix}$, so that

$$\boldsymbol{\mu}^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \boldsymbol{\mu}^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{\Sigma}_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$$

Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$, and consider the linear combinations $\mathbf{AX}^{(1)}$ and $\mathbf{BX}^{(2)}$. Find:

(a) $E(\mathbf{X}^{(1)})$

$E(\mathbf{X}^{(1)})$ is simply the first two entries of $\boldsymbol{\mu}_X$:

$$\boxed{E(\mathbf{X}^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}}$$

(b) $E(\mathbf{AX}^{(1)})$

Using $E(\mathbf{AX}) = \mathbf{A}E(\mathbf{X})$:

$$E(\mathbf{AX}^{(1)}) = \mathbf{A}E(\mathbf{X}^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = (1)(4) + (2)(3) = 4 + 6$$

$$\boxed{E(\mathbf{AX}^{(1)}) = 10}$$

(c) $\mathrm{Cov}(\mathbf{X}^{(1)})$

The covariance matrix of $X^{(1)}$ is simply the upper-left 2×2 block of Σ_X:

$$\boxed{\mathrm{Cov}(\mathbf{X}^{(1)}) = \boldsymbol{\Sigma}_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}$$

(d) $\mathrm{Cov}(\mathbf{AX}^{(1)})$

Using $\mathrm{Cov}(\mathbf{AX}) = \mathbf{A}\,\mathrm{Cov}(\mathbf{X})\,\mathbf{A}'$:

Since

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad A' = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

we get

$$\text{Cov}(\mathbf{AX}^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Multiplying the first two matrices,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix}$$

Then multiplying the resulting product and the last matrix, we get

$$\begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 + 4 = 7$$

$$\boxed{\text{Cov}(\mathbf{AX}^{(1)}) = 7}$$

(e) $E(\mathbf{X}^{(2)})$

This is simply the last two elements of the mean vector

$$\boxed{E(\mathbf{X}^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}}$$

(f) $E(\mathbf{BX}^{(2)})$

Using $E(\mathbf{BX}^{(2)}) = \mathbf{BX}^{(2)}$

So we have,

$$\begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} (1)(2) + (-2)(1) \\ (2)(2) + (-1)(1) \end{bmatrix} = \begin{bmatrix} 2 - 2 \\ 4 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$\boxed{E(\mathbf{BX}^{(2)}) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}}$$

(g) $\text{Cov}(\mathbf{X}^{(2)})$

This is simply the bottom right 2x2 block of the mean vector

$$\boxed{\text{Cov}(\mathbf{X}^{(2)}) = \mathbf{\Sigma}_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}}$$

(h) $\text{Cov}(\mathbf{BX}^{(2)})$

Using $\text{Cov}(\mathbf{BX}) = \mathbf{B}\,\text{Cov}(\mathbf{X})\,\mathbf{B}'$:

We have,

$$\mathbf{B} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad \mathbf{B}' = \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

So,

$$\text{Cov}(\mathbf{B}\mathbf{X}^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

Multiplying the first two matrices, we get

$$\begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} (1)(9) + (-2)(-2) & (1)(-2) + (-2)(4) \\ (2)(9) + (-1)(-2) & (2)(-2) + (-1)(4) \end{bmatrix} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}$$

Now, we multiply the result with the last matrix getting

$$\begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} (13)(1) + (-10)(-2) & (13)(2) + (-10)(-1) \\ (20)(1) + (-8)(-2) & (20)(2) + (-8)(-1) \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$$

Therefore,

$$\boxed{\text{Cov}(\mathbf{B}\mathbf{X}^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}}$$

(i) $\text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)})$

This is simply the upper-right 2x2 block of the mean vector:

$$\boxed{\text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) = \mathbf{\Sigma}_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}}$$

(j) $\text{Cov}(\mathbf{A}\mathbf{X}^{(1)}, \mathbf{B}\mathbf{X}^{(2)})$

Using $\text{Cov}(\mathbf{A}\mathbf{X}^{(1)}, \mathbf{B}\mathbf{X}^{(2)}) = \mathbf{A}\,\text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)})\,\mathbf{B}'$:

Substituting the values we have

$$= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

Multiplying the first two matrices,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} (1)(2) + (2)(1) & (1)(2) + (2)(0) \end{bmatrix} = \begin{bmatrix} 4 & 2 \end{bmatrix}$$

Then multiplying the result with the last matrix,

$$\begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} (4)(1) + (2)(-2) & (4)(2) + (2)(-1) \end{bmatrix} = \begin{bmatrix} 4-4 & 8-2 \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}$$

Therefore,

$$\boxed{\text{Cov}(\mathbf{AX}^{(1)}, \mathbf{BX}^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}}$$

Problem 2.41 **(5 points)**

You are given the random vector $\mathbf{X}' = [X_1, X_2, X_3, X_4]$ with mean vector $\boldsymbol{\mu}'_X = [3, 2, -2, 0]$ and variance-covariance matrix

$$\boldsymbol{\Sigma}_X = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 3\mathbf{I}$$

Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}$$

(a) $E(\mathbf{AX})$, the mean of $\mathbf{AX}$

Using $E(\mathbf{AX}) = \mathbf{A}\boldsymbol{\mu}_X$, with $\boldsymbol{\mu}_X = \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}$:

Therefore,

$$\mathbf{A}\boldsymbol{\mu}_X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} (1)(3) + (-1)(2) + (0)(-2) + (0)(0) \\ (1)(3) + (1)(2) + (-2)(-2) + (0)(0) \\ (1)(3) + (1)(2) + (1)(-2) + (-3)(0) \end{bmatrix}$$

$$= \begin{bmatrix} 3 - 2 \\ 3 + 2 + 4 \\ 3 + 2 - 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

Thus,

$$\boxed{E(\mathbf{AX}) = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}}$$

(b) Cov($\mathbf{AX}$), the variances and covariances of $\mathbf{AX}$

Using Cov($\mathbf{AX}$) $= \mathbf{A\Sigma}_X\mathbf{A}'$, Since $\mathbf{\Sigma}_X = 3I$, we can simplify het equation to $Cov(\mathbf{AX}) = \mathbf{A}(\mathbf{3I})\mathbf{A}' = 3\mathbf{AA}'$.

Let us first compute for $\mathbf{AA}'$ by going through the rows of A:

$$\mathbf{r}_1 = \begin{bmatrix}1 & -1 & 0 & 0\end{bmatrix}, \quad \mathbf{r}_2 = \begin{bmatrix}1 & 1 & -2 & 0\end{bmatrix}, \quad \mathbf{r}_3 = \begin{bmatrix}1 & 1 & 1 & -3\end{bmatrix}$$

$$\mathbf{r}_1\mathbf{r}_1' = 1+1+0+0 = 2, \qquad \mathbf{r}_1\mathbf{r}_2' = 1-1+0+0 = 0, \qquad \mathbf{r}_1\mathbf{r}_3' = 1-1+0+0 = 0$$

$$\mathbf{r}_2\mathbf{r}_2' = 1+1+4+0 = 6, \qquad \mathbf{r}_2\mathbf{r}_3' = 1+1-2+0 = 0, \qquad \mathbf{r}_3\mathbf{r}_3' = 1+1+1+9 = 12$$

So we have,

$$\mathbf{AA}' = \begin{bmatrix}2 & 0 & 0\\0 & 6 & 0\\0 & 0 & 12\end{bmatrix}$$

Now, we multiply by 3,

$$3\mathbf{AA}' = 3\begin{bmatrix}2 & 0 & 0\\0 & 6 & 0\\0 & 0 & 12\end{bmatrix} = \begin{bmatrix}6 & 0 & 0\\0 & 18 & 0\\0 & 0 & 36\end{bmatrix}$$

Hence,

$$\boxed{\text{Cov}(\mathbf{AX}) = \begin{bmatrix}6 & 0 & 0\\0 & 18 & 0\\0 & 0 & 36\end{bmatrix}}$$

Therefore the variances are,

$$\boxed{\text{Cov}(Y_1, Y_2) = 0, \quad \text{Cov}(Y_1, Y_3) = 0\text{Cov}(Y_2, Y_3) = 0}$$

$$\boxed{\mathbf{Var(Y_1) = 6, \quad Var(Y_2) = 18, \quad Var(Y_3) = 36}}$$

(c) Which pairs of linear combinations have zero covariance?

From the previously calculated $\mathbf{Cov}(\mathbf{AX})$ in part (b), we have

$$\text{Cov}(\mathbf{AX}) = \begin{bmatrix}6 & 0 & 0\\0 & 18 & 0\\0 & 0 & 36\end{bmatrix}$$

The off-diagonal entries of $\mathrm{Cov}(\mathbf{AX})$ found in part (b) give the pairwise covariances. It can be easily seen that all off-diagonal entries of $\mathrm{Cov}(\mathbf{AX})$ equal zero, meaning, $\mathrm{Cov}(Y_1, Y_2) = \mathrm{Cov}(Y_1, Y_3) = \mathrm{Cov}(Y_2, Y_3) = 0$,

$$\boxed{\text{every pair among } Y_1, Y_2, Y_3 \text{ has zero covariance.}}$$